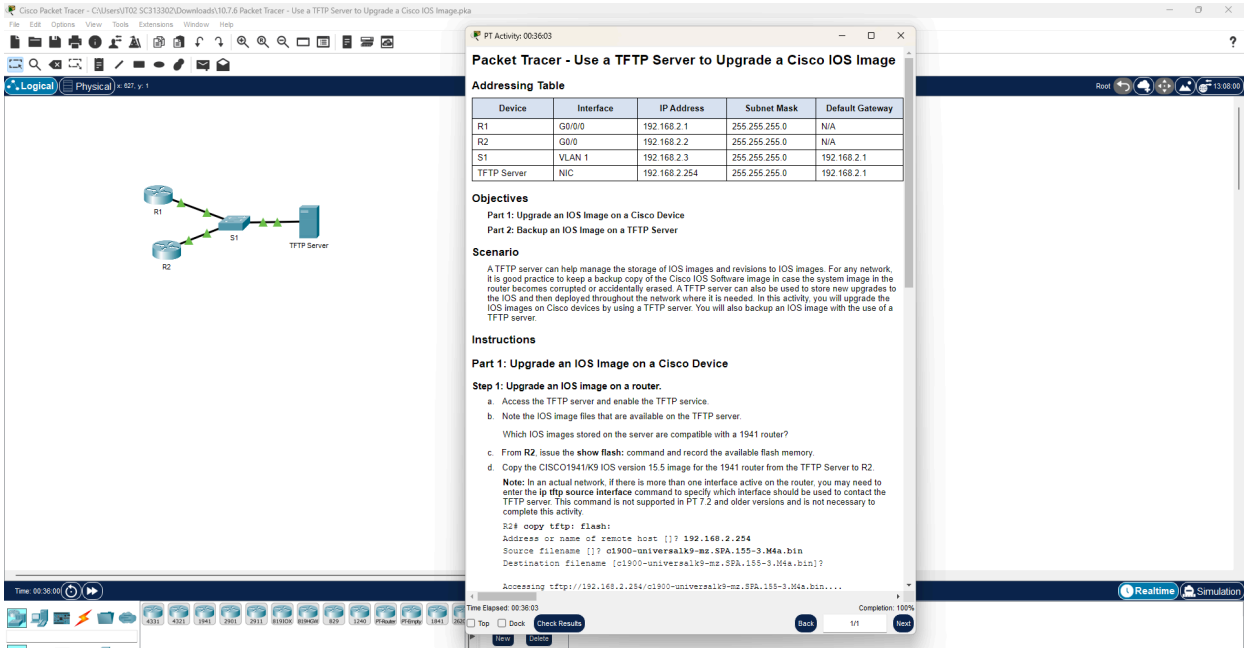


1. Executive Summary & Objectives

Maintaining updated Cisco IOS software images across network infrastructure is vital for securing network devices against vulnerabilities, patching bugs, and enabling new features. Trivial File Transfer Protocol (TFTP) provides a centralized, lightweight mechanism to back up existing firmware images and deploy upgraded software versions across enterprise equipment.

Key Objectives

1. **Upgrade Cisco IOS Image on Devices:** Transfer and install upgraded IOS images onto router **R2** and switch **S1** from a centralized TFTP server, update boot pointers, and verify successful boot execution.
2. **Backup Cisco IOS Image to TFTP Server:** Export the existing system image from router **R1** to the TFTP server to ensure disaster recovery readiness.
3. **Verify System Boot & Readiness:** Validate boot statements using system commands (`show flash:`, `show version`, `boot system`).

2. Network Topology & Addressing Table

The lab topology consists of two routers (**R1**, **R2**), one Catalyst switch (**S1**), and a centralized **TFTP Server** connected on a flat local management subnetwork (`192.168.2.0/24`).

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	192.168.2.1	255.255.255.0	N/A
R2	G0/0	192.168.2.2	255.255.255.0	N/A
S1	VLAN 1	192.168.2.3	255.255.255.0	192.168.2.1
TFTP Server	NIC	192.168.2.254	255.255.255.0	192.168.2.1

Topology Overview

- **Network Segment:** `192.168.2.0/24`
- **Switch Connection: S1** interconnects **R1** (G0/0/0), **R2** (G0/0), and **TFTP Server** (NIC).
- **Gateway Role: R1** acts as default gateway (`192.168.2.1`) for switch **S1** management traffic.

3. Lab Implementation Steps & Execution

Part 1: Upgrade an IOS Image on a Cisco Device

Step 1: Upgrade IOS Image on Router R2

1. **Verify TFTP Service & Compatibility:**
   Access the **TFTP Server** GUI services tab, ensure TFTP is enabled, and review available software image files.
2. **Inspect Current Flash Memory on R2:**
   Issue `show flash:` on **R2** to verify existing image name (`c1900-universalk9-mz.SPA.153-3.M3.bin`) and available storage space (approx. 221,896,412 bytes free).
3. **Copy Upgraded Image from TFTP Server:**
   Transfer the version 15.5 image (`c1900-universalk9-mz.SPA.155-3.M4a.bin`) from `192.168.2.254`:

```
None
R2# copy tftp: flash:
Address or name of remote host []? 192.168.2.254
Source filename []? c1900-universalk9-mz.SPA.155-3.M4a.bin
```

```
Destination filename [c1900-universalk9-mz.SPA.155-3.M4a.bin]?
Accessing tftp://192.168.2.254/c1900-universalk9-mz.SPA.155-3.M4a.bin...
Loading c1900-universalk9-mz.SPA.155-3.M4a.bin from 192.168.2.254:
!!!!!!!!!!!!!!!!!!!
[OK - 33591768 bytes]
```

4. **Configure Boot Variable & Reload:**
   Set **R2** to boot the newly copied image, save configuration, and reboot:

```
None
R2(config)# boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin
R2(config)# exit
R2# copy running-config startup-config
R2# reload
```

5. **Verify Upgraded IOS Load:**
   After reload, execute `show version` to verify system image file path: `System image file is "flash0:c1900-universalk9-mz.SPA.155-3.M4a.bin"` (Version 15.5(3)M4a).

Step 2: Upgrade IOS Image on Switch S1

1. **Transfer Image to Switch S1:**
   Transfer switch firmware image `c2960-lanbasek9-mz.150-2.SE4.bin` from `192.168.2.254`:

```
None
S1# copy tftp: flash:
Address or name of remote host []? 192.168.2.254
Source filename []? c2960-lanbasek9-mz.150-2.SE4.bin
Destination filename [c2960-lanbasek9-mz.150-2.SE4.bin]?
```

2. **Configure System Boot Path & Reload:**
   Set system boot command, write memory, and restart switch:

```
None
S1(config)# boot system flash:c2960-lanbasek9-mz.150-2.SE4.bin
S1(config)# exit
S1# copy running-config startup-config
S1# reload
```

3. **Verify Upgraded Image:**
   Verify active image using `show version`.

Part 2: Backup an IOS Image to a TFTP Server

1. **Check Flash Contents on R1:**
   Execute `show flash:` on **R1** to obtain exact image filename: `isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin`.
2. **Transfer Image from R1 to TFTP Server:**
   Execute image backup to server address `192.168.2.254`:

```
None
R1# copy flash: tftp:
Source filename []? isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
Address or name of remote host []? 192.168.2.254
Destination filename [isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin]?
Writing isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin...
!!!!!!!!!!!!!!!!!!!
[OK - 486899872 bytes]
```

3. **Verify Backup on TFTP Server:**
   Open TFTP Server directory and verify `isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin` appears in file repository.

4. Complete Device Configurations

Router R1 Configuration

```
None
hostname R1
!
interface GigabitEthernet0/0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
!
ip forward-protocol nd
!
line con 0
 line vty 0 4
 login
!
end
```

Router R2 Configuration

```
None
hostname R2
!
boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin
!
interface GigabitEthernet0/0
 ip address 192.168.2.2 255.255.255.0
 no shutdown
!
ip forward-protocol nd
!
line con 0
 line vty 0 4
 login
!
end
```

Switch S1 Configuration

```
None
hostname S1
!
boot system flash:c2960-lanbasek9-mz.150-2.SE4.bin
!
interface Vlan1
 ip address 192.168.2.3 255.255.255.0
 no shutdown
!
ip default-gateway 192.168.2.1
!
line con 0
 line vty 0 4
 login
!
end
```

5. Lab Questions and Detailed Answers

Step 1b Question

Question: Which IOS images stored on the server are compatible with a 1941 router?

- **Answer:** `c1900-universalk9-mz.SPA.155-3.M4a.bin` (and `c1900-universalk9-mz.SPA.150-1.M4.bin`). Cisco 1941 routers belong to the Cisco 1900 Series Integrated Services Router (ISR) architecture, requiring images designated with the `c1900` hardware platform prefix.

Step 1c Question

Question: Issue the `show flash:` command and record the available flash memory on R2.

- **Answer:** Initial flash memory allocation on **R2**:
  - Total capacity: ~255,426,560 bytes (~256 MB)
  - Used memory: ~33,530,148 bytes (Original IOS image)
  - Free/Available space: **221,896,412 bytes** (~221.9 MB)

Step 1e Question

Question: How many IOS images are located in flash on R2 after copying the new image?

- **Answer: Two (2) IOS images** are stored in flash memory:
    a. `c1900-universalk9-mz.SPA.153-3.M3.bin` (Original pre-installed image)
    b. `c1900-universalk9-mz.SPA.155-3.M4a.bin` (Newly upgraded image copied via TFTP)

6. Technical Analysis, Edge Cases & Troubleshooting

Source Interface Requirement (`ip tftp source-interface`)

In production enterprise networks with multi-interface routers, TFTP request packets originate from the egress interface closest to the server by default. If access control lists (ACLs) or firewall rules restrict traffic to specific management interfaces, issue the source interface override:

```
None
R2(config)# ip tftp source-interface GigabitEthernet0/0
```

Edge Case Scenarios & Recovery Strategies

1. **TFTP File Transfer Failure / Timeout (`.....` followed by failure):**
    - *Root Cause:* IP address misconfiguration, VLAN misassignment, or missing default gateway on switch.
    - *Resolution:* Ping server (`ping 192.168.2.254`) to verify Layer 3 connectivity before starting file transfers.
2. **Insufficient Flash Memory Space:**
    - *Root Cause:* Older routers with small flash chips cannot hold two large system images simultaneously.
    - *Resolution:* Delete outdated images using `delete flash:<filename>` before copying new ones. Always verify that free flash bytes exceed destination image size.
3. **Corrupted Image / ROMMON Recovery:**
    - *Root Cause:* Aborted file transfer, corrupt download, or bad checksum.
    - *Resolution:* If a router enters `rommon 1>` prompt upon reload, restore system via TFTP download within ROMMON mode:

```
None
rommon 1> IP_ADDRESS=192.168.2.2
rommon 2> IP_SUBNET_MASK=255.255.255.0
rommon 3> DEFAULT_GATEWAY=192.168.2.1
rommon 4> TFTP_SERVER=192.168.2.254
rommon 5> TFTP_FILE=c1900-universalk9-mz.SPA.155-3.M4a.bin
rommon 6> tftpdnld
```

4. **Image Verification via MD5 Hash:**
    - Validate image integrity on CLI before rebooting:

```
None
R2# verify /md5 flash:c1900-universalk9-mz.SPA.155-3.M4a.bin
```

7. Real-World Production Suggestions & Alternatives

- **Secure Protocols (SFTP / SCP):** TFTP transmits files in plain text over UDP port 69 without authentication. Secure Copy Protocol (SCP) or SFTP should replace TFTP in enterprise networks to prevent image tampering and eavesdropping.
- **Cisco DNA Center / Cisco Prime Infrastructure:** Automated network management tools replace manual CLI file transfers with centralized batch image distribution and automated compliance verification.

8. Grading Rubric & Self-Reflection

Criteria	Target Requirement	Evaluation / Status
R2 Upgrade	Upgrade to `c1900-155-3` image & update boot system path	**PASS** (Version 15.5 verified)
S1 Upgrade	Upgrade to `c2960-150-2` image & set boot path	**PASS** (Boot path verified)

Criteria	Target Requirement	Evaluation / Status
R1 Backup	Export `isr4300` image to TFTP server `192.168.2.254`	**PASS** (Image present on TFTP)
Documentation	Complete CLI logs, topology table, and question answers	**PASS** (All items answered)

Self-Reflection

Completing this lab provided practical experience in centralized Cisco IOS software deployment and backup operations. Configuring explicit `boot system` statements ensures predictable system boots after upgrades and prevents routers from falling back to older images or entering boot loops.